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DIBRIS

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

COGNITIVE ARCHITECTURES FOR ROBOTICS

FINAL PROJECT

A-to-B Navigation with G1 EDU in Gazebo

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1. Introduction

This project develops a standalone A-to-B indoor navigation benchmark for the Unitree G1 EDU robot in Gazebo using ROS 2 Humble and Nav2. The robot is modeled with 29 degrees of freedom and uses a Livox MID-360 LiDAR and an Intel RealSense D435i depth camera.

The system performs navigation on a known map with simulator-based localization, global path planning, local obstacle avoidance, and recovery behaviors. Three local planners—DWB, MPPI, and RPP—are compared in corridor, doorway, clutter, dynamic blockage, and maze scenarios.

Performance is evaluated using success rate, completion time, path efficiency, collision events, recovery count, and final goal error. The main objective is to identify the most suitable planner configuration for reliable indoor navigation of the Unitree G1 platform.

2. Project Objective and Scope

The goal was not simply to demonstrate one successful navigation run. The project was designed as a controlled benchmark that could answer a practical question:

Which Nav2 local planner is most suitable for a humanoid-sized robot abstraction moving through structured indoor environments?

The implemented pipeline included a known map, simulator-provided baseline localization, global route generation, local obstacle-aware control, execution monitoring, recovery behavior, logging, and automatic metric generation. The Unitree G1 abstraction used a MID-360-style LiDAR and a D435i-style depth camera. The LiDAR point cloud was adapted to a 2D LaserScan stream for Nav2.

Implemented components

Component	Implementation
Simulation	Gazebo with the Unitree G1 EDU abstraction
Global planning	ThetaStar for all planner profiles
Local Planner	DWB, MPPI, and Regulated Pure Pursuit
Localization	Simulator baseline map-to-odom and odometry
Recovery	Nav2 behavior-tree recovery after progress failure
Evaluation	CSV telemetry, summary metrics, and plots

3. Cognitive and Software Architecture

The system is described as a **hybrid reactive–deliberative architecture**. Its main navigation chain follows the hierarchical Sense–Plan/Reasoning–Act organization. Sensor and localization information continuously update the navigation state, while the mission is represented as a `GO-TO(start, goal)` task. The Robot Execution Manager, realized primarily through the Nav2 Behavior Tree, coordinates global path planning, local motion planning and control, replanning, and recovery operations. Execution monitoring compares the active path and expected progress with the observed robot pose, motion, and environment state. During normal progress, the system returns a

Continue outcome and maintains execution of the current plan. If the robot reaches the goal, the task reports **Success**. If the active path becomes invalid because of an environmental change, **Replan** can be requested while preserving the original goal. If the robot loses progress or becomes stuck, Nav2 recovery behaviors may be triggered. If recovery cannot restore progress, the task may ultimately report Failure.

At the ROS 2 communication level, the simulated 3D LiDAR publishes `sensor_msgs/msg/PointCloud2` data on `/livox/points`. The `pointcloud_to_laserscan` adapter converts this raw three-dimensional point cloud into an adapted two-dimensional `sensor_msgs/msg/LaserScan` stream on `/scan`.

The Nav2 global and local costmaps are both configured to receive the adapted LaserScan, but they use it in different ways. The global costmap combines the known static map with a `/scan`-based obstacle layer to account for obstacles affecting the planned route. The local costmap uses `/scan` for continuously updated detection of nearby obstacles required by the local planner.

The adapted LaserScan on `/scan`, rather than the raw PointCloud2 stream, is also supplied directly to the Braitenberg reflex. This creates a fast Sense-to-Act pathway that can respond to an immediate obstacle without waiting for global replanning.

In the present benchmark, the LiDAR-derived LaserScan was sufficient for two-dimensional navigation and obstacle avoidance. The RGB-D camera is therefore retained as an available sensor for future extensions such as object recognition, semantic perception, close-range manipulation, or sensor fusion.

Localization and robot state are represented through the ROS 2 TF chain:

```
map → odom → base_link
```

A baseline transform provides the `map → odom` relation, while the simulated robot publishes odometry on `/odom` and the corresponding `odom → base_link` transform. The `/odom` topic supplies the observed motion state used by Nav2 control and progress checking, and it is also recorded by the benchmark logger. Together with TF and costmap observations, this information closes the execution-monitoring loop by allowing actual robot progress to be evaluated against the active goal and path.

Navigation goals are sent through the Nav2 `/navigate_to_pose` **action**, using `nav2_msgs/action/NavigateToPose`. An action is appropriate because navigation is a long-running operation that provides feedback, returns success or failure, and can be cancelled.

The `benchmark_runner` externally orchestrates each experiment and maps to the **Task/Goal Manager** role shown in the Plan/Reasoning layer. It loads the selected scenario, reads the start and goal configuration, sends the navigation goal through the action interface, waits for the result, and records the final task status. It should therefore be understood as the software component that realizes the Task/Goal Manager function for the benchmark, rather than as an additional cognitive layer.

Within the normal deliberative path, the Global Path Planner uses ThetaStar to generate a route toward the goal. The selected Local Planner—DWB, MPPI, or RPP—then follows the global route and produces the normal velocity command.

The Local Planner generates the normal velocity command on `~/nav_cmd_vel`. The subsumption coordinator normally forwards this command but temporarily selects `~/reflex_cmd_vel` when the Braitenberg reflex is active. The selected command is published on `~/cmd_vel` and sent to the robot.

The reactive safety pathway uses the following ROS 2 topics:

- `/scan` provides the adapted obstacle measurements to the Braitenberg reflex.
- `/reflex_cmd_vel` carries the safe velocity command generated by the reflex.
- `/reflex_active` indicates whether the reflex should temporarily override normal navigation.
- `/reflex_min_range` reports the minimum detected obstacle distance for monitoring and debugging.
- `/nav_cmd_vel` carries the normal velocity command generated by Nav2.
- `/cmd_vel` carries the final selected command sent to the locomotion interface.
- `/control_mode` reports whether the system is operating in normal navigation, reflex override, or idle mode.

The subsumption coordinator subscribes to `/nav_cmd_vel`, `/reflex_cmd_vel`, and `/reflex_active`. During normal operation, it forwards the Nav2 velocity command. When an obstacle activates the reflex and a valid reflex command is available, the coordinator temporarily selects the reflex command to reduce collision risk. The resulting `geometry_msgs/msg/Twist` command is published on `/cmd_vel` and consumed by the Gazebo locomotion interface.

Execution monitoring and experiment recording are distributed across the Nav2 Behavior Tree, its progress-checking and recovery mechanisms, the benchmark runner, and the benchmark logger rather than being implemented as one independent ROS 2 node. The benchmark logger subscribes to `/odom`, `/cmd_vel`, `/scan`, and `/reflex_active`. These streams are used to record the executed trajectory, commanded motion, obstacle clearance, stopped time, and reflex activation.

The ROS 2 communication mechanisms therefore serve different architectural purposes:

- **Topics** carry continuous sensor, state, monitoring, and velocity information.
- The **NavigateToPose action** manages the long-running navigation task, its feedback, and its final result.
- **Services** perform short discrete operations, such as enabling logging or modifying the simulated environment.
- **TF transforms** maintain the spatial relationships required by localization, costmaps, planning, and control.

The final architecture preserves the three core course layers—Sense, Plan/Reasoning, and Act—while adding a fast bio-inspired safety pathway. The normal Nav2 chain provides deliberate, goal-directed navigation and can continue, replan, recover, or report success according to the observed execution state. In parallel, the Braitenberg reflex maps immediate adapted LaserScan measurements directly to a safe velocity response. The subsumption coordinator arbitrates between the normal command and the reflex override, making the complete system a hybrid reactive–deliberative cognitive architecture.

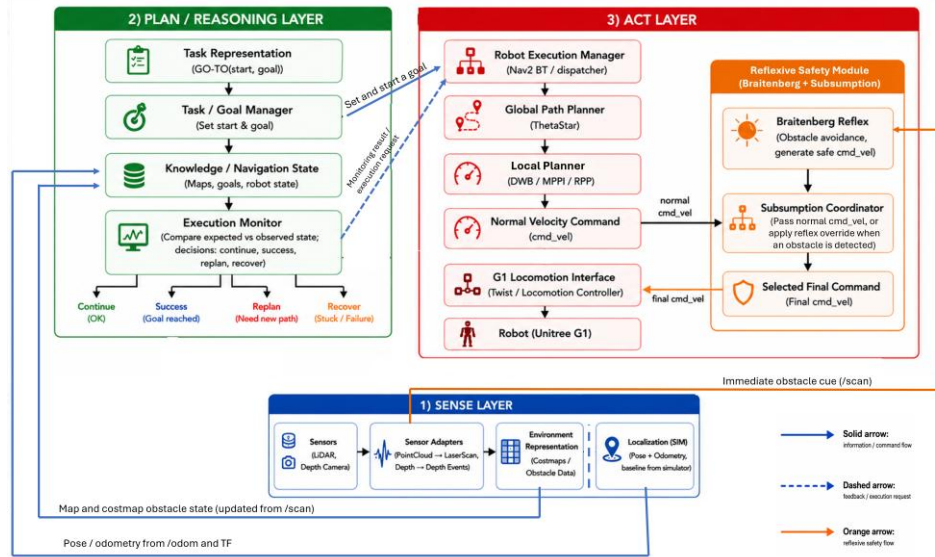


Figure 1. Hybrid reactive–deliberative navigation architecture of the project.

3. Experimental Design

A fair comparison required the environment and global planning conditions to remain fixed. All three local planners used the same maps, start and goal poses, robot footprint, costmaps, velocity limits, goal tolerance, startup procedure, timeout, and ThetaStar global planner. Only the local planner profile changed.

The benchmark matrix contained five scenarios, three planner profiles, and three repetitions, for a total of 45 runs. The scenarios were selected to increase difficulty gradually from basic timing to dynamic rerouting and high-curvature navigation.

Scenario	Purpose	Main challenge
Corridor	Baseline timing	Straight, open navigation
Doorway	Narrow-passage test	Footprint and alignment
Clutter	Static obstacle adaptation	Local trajectory selection
House dynamic	Dynamic rerouting	Route blockage during execution
Maze	Complex route following	High curvature and constrained geometry

Metrics and evaluation priority

The report treats success rate and completion time as the primary criteria, because they directly answer whether the task was completed and how efficiently it was completed. Final goal error, recovery count, minimum clearance, path efficiency, and collision proxy events were used as supporting measures.

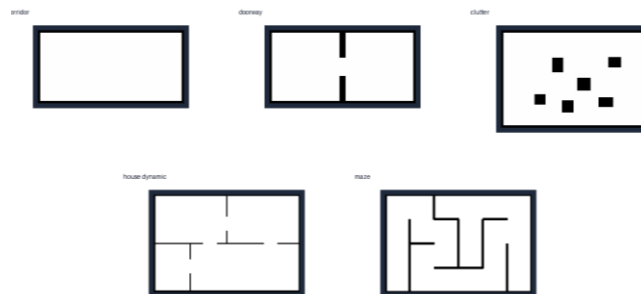


Figure 2. Five indoor tasks stress different navigation behaviors.

4. Primary Results: Success and Completion Time

The main result is clear. All planners completed the three easier scenarios. The separation appeared in house_dynamic and maze: MPPI and RPP completed every run, while DWB failed all repetitions in both scenarios. Across the full benchmark, this corresponds to 100% overall success for MPPI and RPP and 60% for DWB.

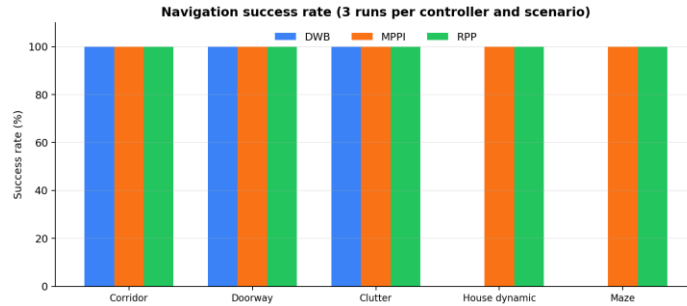


Figure 3. Success rate by planner and scenario. DWB failed consistently in the two hardest tasks.

Completion time reinforces the same conclusion. In the complex successful cases, MPPI was faster than RPP: approximately 69.1 s versus 76.9 s in house_dynamic, and 136.1 s versus 150.1 s in maze. DWB approached the run timeout in both failed scenarios, with mean durations of approximately 185.0 s and 196.1 s.

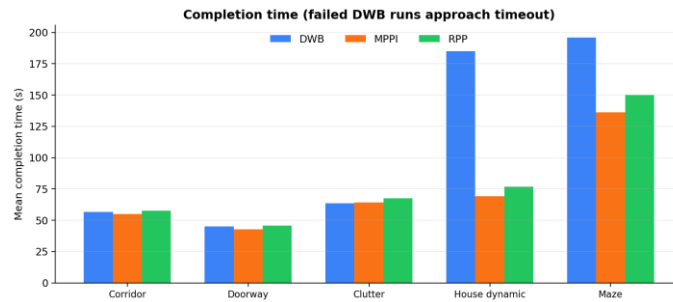


Figure 4. Mean completion time. The high DWB values in the hard scenarios correspond to failed runs approaching timeout.

Interpretation

MPPI did not only succeed more reliably; it also preserved competitive completion time in complex geometry. This supports the expected advantage of predictive, sampled control when local decisions become difficult.

5. Recovery, Accuracy, and Safety

DWB did not fail because of a recorded obstacle-contact event. Its main failure mode in the difficult scenarios was loss of progress in sharp turns and high-curvature sections. It triggered an average of two recovery actions in house_dynamic and three in maze. However, the available recovery sequence did not move the robot into a sufficiently different state to escape the local minimum and resume navigation. This is a valid benchmark finding: recovery was triggered correctly, but it was not effective for the specific failure condition encountered by DWB.

MPPI achieved the smallest final goal error in every scenario. RPP also completed every task, although it generally stopped with a slightly larger residual error. No collision-proxy events were recorded in the final aggregate results. However, this metric is based on obstacle proximity and should not be interpreted as physical-contact ground truth.

Planner	Overall success	Hard-case recovery	Braitenberg reflex activation	Goal accuracy	Assessment
DWB	60%	2 in house_dynamic; 3 in maze	1 activation per run in house_dynamic	Large error on failed runs	Useful baseline; weak in hard cases
MPPI	100%	No Nav2 recovery required	1 activation per run in house_dynamic	Best in all scenarios	Best overall profile
RPP	100%	No Nav2 recovery required	1 activation per run in house_dynamic	Reliable, slightly less precise	Robust but conservative

Recovery interpretation

Recovery should not be confused with normal obstacle avoidance or reactive safety. In the house_dynamic scenario, a door-like obstacle suddenly appeared and blocked the current route. The Braitenberg-inspired reflex responded to the immediate obstacle cue and temporarily modified the velocity command, redirecting the robot away from the obstacle and preventing an unsafe approach.

The updated obstacle information was then incorporated into the navigation system. Global replanning generated an alternative route, and the local planners followed it. DWB, MPPI, and RPP were all able to respond correctly to the dynamic blockage itself. Therefore, the recovery events recorded for DWB were not caused directly by the appearance of the door. They occurred later, when DWB encountered sharp turns or high-curvature sections and could no longer maintain sufficient progress.

In these cases, Nav2 correctly detected the loss of progress and activated recovery. However, the available recovery behaviors did not produce enough positional change to move the robot out of the local minimum. As a result, DWB could return to nearly the same configuration and fail again.

Therefore, the three mechanisms had distinct roles: the Braitenberg reflex provided immediate local safety, the global planner generated an alternative route around the blockage, and recovery was activated only when normal navigation later failed to make progress.

6. Discussion

The results support a planner ranking for this specific simulation configuration, not a universal claim. MPPI was the strongest overall choice because it combined complete task success, competitive time, and the smallest final error. RPP was a strong practical alternative: it was safe and reliable, but its regulated behavior was slower and slightly less precise. DWB remained valuable as a transparent baseline, but its tuning and recovery interaction were not sufficient for the dynamic and maze tasks.

From a COGAR perspective, the project is more than a planner comparison. It demonstrates how symbolic task intent, metric planning, continuous control, execution monitoring, recovery, and fast reactive behavior can be integrated into a reproducible ROS 2 system. The Braitenberg-inspired reflex was exercised in the 'house_dynamic' scenario, where it provided an immediate sensory-motor safety response to the suddenly appearing obstacle. Its role was to modify the short-term velocity command, while global replanning and local planner remained responsible for generating and following the alternative route.

7. Limitations and Future Work

- The evaluation is simulation-only and uses simulator-provided baseline localization.
- Each planner-scenario pair was repeated three times; more repetitions would strengthen statistical confidence.
- The locomotion model was intentionally simplified because the main objective of the project was navigation planning, obstacle avoidance, and planner benchmarking rather than full-body humanoid motion control. Realistic locomotion for a 29-DOF humanoid would require whole-body dynamics, balance control, gait generation, and significantly greater computational resources.
- Future work should include real G1 trials, moving-human scenarios, parameter sensitivity analysis, and failure-specific recovery behaviors.

8. Conclusion

This work produced a complete and repeatable A-to-B navigation benchmark for the Unitree G1 EDU abstraction in Gazebo. By fixing ThetaStar global planner and varying only the local planner, the experiments isolated meaningful differences between DWB, MPPI, and RPP. The primary criteria - success and completion time - identify MPPI as the strongest overall profile in the tested scenarios. RPP was equally reliable but more conservative, while DWB failed in the two hardest tasks despite triggering recovery.

The final contribution is therefore both engineering and cognitive: a reproducible ROS 2 benchmark, and a clear mapping from COGAR concepts to a working hybrid architecture with hierarchical planning, execution monitoring, recovery, and reactive safety.

Final takeaway

MPPI is the recommended planner for the final G1 benchmark configuration. RPP is a robust alternative. DWB should remain as the baseline and as evidence of where the current planner/recovery combination loses progress.

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